

Chapter 11 – Google Drive, Shared Drives & Collaboration

Learning Objectives

After completing this chapter, you should be able to:

- Understand My Drive and Shared Drives.
 - Explain ownership and file permissions.
 - Differentiate Drive roles.
 - Design enterprise Shared Drives.
 - Configure external sharing.
 - Explain trusted domains.
 - Understand Drive audit logging.
 - Troubleshoot common Drive access issues.
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1. Business Problem

Imagine a company stores all HR policies, contracts, onboarding documents, and payroll spreadsheets in the HR Director's personal Google Drive.

One day, the HR Director resigns.

What happens?

Possible issues:

- Files belong to the HR Director.
- Employees lose access if the account is deleted.
- Ownership transfer becomes necessary.
- Business operations may stop.

This is why organizations use **Shared Drives** instead of personal drives for team-owned data.

2. Google Drive Overview

Google Drive is Google's cloud storage service.

Users can store:

- Documents
- PDFs
- Images
- Videos
- Presentations
- Spreadsheets
- Forms

Drive supports:

- Collaboration
- Sharing
- Version history
- Access control
- Audit logging

3. My Drive

My Drive is a user's personal storage area.

Characteristics:

- User owns the files.
- User controls sharing.
- Files are associated with the user's account.
- Ownership may need to be transferred if the user leaves.

Example:

rahul@company.com

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My Drive

- └ Resume.docx
- └ Notes
- └ Personal Projects
- └ Training

Suitable for:

- Personal work
- Draft documents
- Temporary files

4. Shared Drives

Shared Drives are designed for teams.

Key principle:

The organization owns the files—not an individual user.

Characteristics:

- No individual file owner.
- Membership-based access.
- Files remain even if members leave.
- Better for long-term collaboration.

Example:

HR Shared Drive

- └ Employee Handbook
- └ Payroll Policies
- └ Recruitment
- └ Onboarding

If the HR Director leaves,

the files remain available to the HR team.

This is exactly the scenario we discussed.

My Drive vs Shared Drive

My Drive	Shared Drive
Individual ownership	Organization ownership

My Drive	Shared Drive
Personal workspace	Team workspace
Files may require ownership transfer	Files remain with the organization
Suitable for personal work	Suitable for departmental data

5. Enterprise Example

Company

Contoso Ltd.

Departments

- HR
- Finance
- Engineering
- Sales

Each department receives its own Shared Drive.

Shared Drives

└─ HR
└─ Finance
└─ Engineering
└─ Sales

Benefits:

- Centralized collaboration
- Easier administration
- Simplified permissions
- No ownership issues

6. Shared Drive Roles

Shared Drives provide several permission levels.

Role	Capabilities
Manager	Full administrative control, including membership management
Content Manager	Create, edit, move, and delete content (cannot manage members)
Contributor	Create and edit files
Commenter	Comment only
Viewer	Read-only access

7. Content Manager

This was one of our interview questions.

Content Manager can:

- Upload files
- Edit files
- Move files within the Shared Drive
- Delete content
- Organize folders

Cannot:

- Add or remove members
- Change membership permissions

Typical use case:

Department Leads

They manage content without administering membership.

Manager vs Content Manager

Manager	Content Manager
Manage members	No
Manage permissions	No
Delete content	Yes

Manager	Content Manager
Move content	Yes
Full administrative control	Limited to content management

8. File Sharing

Google Drive supports multiple sharing options.

Examples:

- Private
- Specific users
- Groups
- Entire organization
- External users
- Public (if allowed)

Sharing should align with organizational policies.

9. Trusted Domains

Suppose a law firm works closely with your company.

Requirement:

- Allow sharing with the law firm.
- Block sharing with personal Gmail accounts.

Solution:

1. Add the law firm's domain as a trusted domain.
2. Allow external sharing only with trusted domains.
3. Block sharing with gmail.com and other untrusted domains.

This is the exact answer you gave during our practice session.

Enterprise Example

Allowed:

company.com

lawfirm.com

Blocked:

gmail.com

yahoo.com

outlook.com

Employees can collaborate securely with approved partners.

10. Link Sharing

Organizations can control link sharing.

Typical options:

- Restricted
- Organization only
- Trusted domains
- Anyone with the link (if allowed)

Most enterprises disable public link sharing unless there is a business requirement.

11. Drive Audit Logs

Every significant Drive activity can be logged.

Examples:

- File created
- File deleted
- Sharing changed
- Downloaded
- Ownership changed
- Permission modified

Audit logs help administrators investigate incidents and demonstrate compliance.

12. Investigation Tool

Available with Enterprise editions (for example, Enterprise Plus),

the Investigation Tool allows administrators to:

- Search Drive events
- Investigate suspicious activity
- Correlate events across services
- Take actions directly from the investigation

Example:

A confidential file was shared externally.

The administrator can:

- Identify who shared it.
- Determine when it happened.
- Review permission changes.
- Remove external access if appropriate.

13. Drive Security

Common security controls include:

- External sharing restrictions
- Trusted domains
- Data Loss Prevention (DLP)
- Context-Aware Access
- Drive labels (edition-dependent)
- Audit logging

These controls help protect sensitive information.

14. Common Drive Issues

User Lost Access

Possible causes:

- Removed from Shared Drive
- Group membership changed
- Sharing permission removed

Resolution:

Verify membership and permissions.

Files Missing After Employee Leaves

Possible cause:

Files were stored in My Drive.

Resolution:

Use Shared Drives for team-owned content.

External Sharing Blocked

Possible causes:

- Domain not trusted
- Organization policy
- DLP rule

Resolution:

Review external sharing settings and trusted domain configuration.

Cannot Delete Files

Possible cause:

Insufficient Shared Drive role.

Resolution:

Review the user's role.

15. Troubleshooting Framework

When users report Drive issues,

ask:

1. Is the file in My Drive or a Shared Drive?
2. Does the user still have permission?
3. Is the user a member of the Shared Drive?
4. Has external sharing been restricted?
5. Is a DLP or security policy blocking the action?
6. Review Drive audit logs.
7. Use the Investigation Tool if needed.

Following a structured approach avoids unnecessary permission changes.

Real Enterprise Scenario

The HR Director resigns.

All HR documents are stored in:

My Drive.

Problem:

The organization must transfer ownership before deleting the account.

Better solution:

Store departmental documents in an HR Shared Drive.

The organization retains ownership,
and operations continue without interruption.

This was exactly the scenario we discussed.

Best Practices

- ✓ Use Shared Drives for team-owned data.
 - ✓ Keep personal files in My Drive.
 - ✓ Apply the Principle of Least Privilege.
 - ✓ Use Groups for permission management.
 - ✓ Configure trusted domains.
 - ✓ Review sharing settings regularly.
 - ✓ Monitor Drive audit logs.
 - ✓ Use the Investigation Tool for security investigations.
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Common Mistakes

- ✗ Storing departmental data in My Drive.
 - ✗ Granting Manager role unnecessarily.
 - ✗ Allowing unrestricted external sharing.
 - ✗ Ignoring audit logs.
 - ✗ Sharing directly with individuals instead of Groups when team access is required.
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Interview Questions

1. What is the difference between My Drive and Shared Drives?

2. Why are Shared Drives recommended for enterprise teams?
 3. What can a Content Manager do?
 4. How does a Manager differ from a Content Manager?
 5. How would you allow collaboration with a law firm while blocking Gmail accounts?
 6. What happens if the HR Director owns all department files and leaves the company?
 7. How would you investigate who shared a confidential document externally?
 8. Which edition provides the Investigation Tool?
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Key Takeaways

- My Drive is owned by the individual user.
 - Shared Drives are owned by the organization.
 - Shared Drives prevent ownership issues when employees leave.
 - Content Managers manage content but not membership.
 - Trusted domains enable secure collaboration with approved external organizations.
 - Drive audit logs and the Investigation Tool are essential for security investigations.
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Chapter Summary

In this chapter you learned:

- ✓ My Drive.
- ✓ Shared Drives.
- ✓ Drive ownership.
- ✓ Shared Drive roles.
- ✓ Content Manager.
- ✓ External sharing.
- ✓ Trusted domains.

✓ Drive audit logs.

✓ Investigation Tool.

✓ Enterprise collaboration best practices.

The next chapter will cover **Google Workspace Security & Compliance**, including **2-Step Verification (2SV), Security Center, Data Loss Prevention (DLP), Google Vault, Context-Aware Access, Endpoint Management, Security Health, Alert Center, and compliance workflows**. This chapter will tie together many of the security concepts we've already practiced, such as Context-Aware Access policies, login challenges, investigations, and enterprise security operations.